

Homework _num_2

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Question _num_1

- a. Valid Operation
- b. Valid Operation
- c. Key Constraint ($\text{Key_num}_4 \text{is already taken}$) $\text{KeyConstraint}, \text{NotNullConstraint}(\text{Key cannot be null})$
- d. Valid Operation
- e. Valid Operation
- f. Integrity Constraint (Did not update ORKS_ON \end{verb} or \begin{verb} DEPENDANT \end{verb})
- g. Integrity Constraint (Did not update ORK_ON \end{verb})
- h. Valid Operation
- i. Integrity Constraint (Did not update ORKS_ON \end{verb})
- j. Valid Operation

Question _num_2

Grader note: the rho (ρ) is too small in L^AT_EX to be distinguished from a subscript, so instead P is used in it's place.

- a. $\Pi_{\text{Fname}, \text{Minit}, \text{Lname}}(\sigma_{\text{Dno} = '5' \wedge \text{hours} = > 10}(\text{roject} \end{verb} \bowtie _ \{ \text{Dname} = 'roject' \end{verb} \end{verb}))$
- b. $\Pi_{\text{Fname} = \text{Dependent_Name}}(\text{mployee} \end{verb} \bowtie \begin{verb} \text{Dependent} \end{verb}) \$$
- c. $\Pi_{\text{Fname}, \text{LName}}(\text{mployee} \end{verb} \div \Pi_{\{ \text{Super_SSN} \}} (\sigma_{_ \{ \text{Fname} = 'Franklin' \end{verb} \end{verb}))$
- d. $\Pi_{\text{Lname}, \text{Sum_Hours}}(s_{\text{name}} \mathcal{F}_{\text{sum hours}}(\text{ROJECT} \end{verb} \bowtie \rho_{\{ \text{ESSN} = \text{SSN} \}} \begin{verb} \end{verb}))$
- e. $\text{mployee1} \end{verb} \leftarrow \begin{verb} \text{mployee} \bowtie \rho_{\text{ESSN} = \text{SSN}} \text{ork_On} \end{verb} \$ \Pi_{\text{Lname}, \text{Sum_Hours}}$

- f. $\Pi_{Lname}(\Pi_{Lname, SSN} \text{employee} - \Pi_{\text{Lname, SSN}}(\text{employee} \bowtie \rho_{ESSN = SSN} \text{orks_On}))$
- g. $\Pi_{Fname, Average Salary}(\text{DName} \mathcal{F}_{average salary} P_{Dno = Dnumber} \text{employee} \bowtie (\text{Department} \bowtie \rho_{Dname = Dnumber} \text{works_On}))$
- h. $\text{Sex} \mathcal{F}_{average salary}(\sigma_{sex = 'female'} \text{employee})$
- i. $\text{Department1} \rightarrow \rho_{\text{Number} = Dnumber} \text{Department} \bowtie \rho_{\text{Dname} = Dnumber} \text{works_On}$
- j. $\Pi_{Lname}(\text{employee} \bowtie \rho_{\text{SSN} = MGR_SSN} \text{Department}) - \Pi_{Lname} \text{employee}$

Question num₃

- $\text{ust\#} \rightarrow \rho_{\text{ORDER}} \text{ORDER}$
- $\text{rder} \rightarrow \rho_{\text{RDER_ITEM, SHIPMENT}} \text{RDER_ITEM, SHIPMENT}$
- $\text{tem\#} \rightarrow \rho_{\text{ORDER}} \text{ORDER}, \rho_{\text{ITEM}} \text{ITEM}$
- $\text{arehouse\#} \rightarrow \rho_{\text{SHIPMENT}} \text{SHIPMENT}$

Assumptions

- Table headers with the same name are not unique.